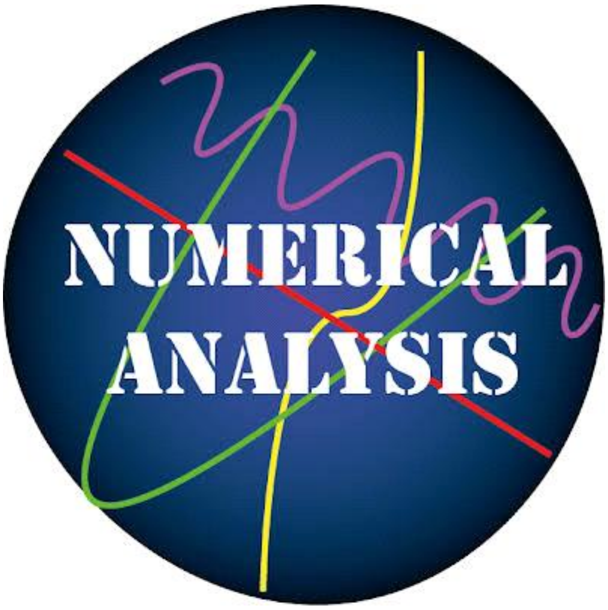
Lec-01

Friday, 14 August 2026 6:43 pm



Numerical analysis is a branch of mathematics that solves continuous problems using numeric approximation. It involves designing methods that give approximate but accurate numeric solutions, which is useful in cases where the exact solution is impossible or prohibitively expensive to calculate. Numerical analysis also involves characterizing the convergence, accuracy, stability, and computational complexity of these methods.

- Interpolation, extrapolation, and regression
- Differentiation and integration
- Linear systems of equations
- Eigenvalues and singular values
- Ordinary differential equations (ODEs)
- Partial differential equations (PDEs)

Topics covered
Calculus review and error analysis
Bisection method and error analysis
Fixed point iterations
Newton's method and error analysis for iterative methods
Interpolation, Lagrange polynomial, Neville's method
Divided difference, Hermite polynomial
Cubic Spline and parametric curves
Numerical differentiation,Richardson's extrapolation
Numerical integration: Trapezoidal and Simpson's rules
Numerical quadratures
Initial value problem, Euler's mathod, higher order Taylor's methods
Runge-Kutta methods and error analysis
Norms of vectors and matrices, and eigenvalues
Numerical approaches to system of linear equations

Gottfried Wilhelm Leibniz was the first to state clearly the rules of calculus.

Isaac Newton developed the use of calculus in his laws of motion and universal gravitation.

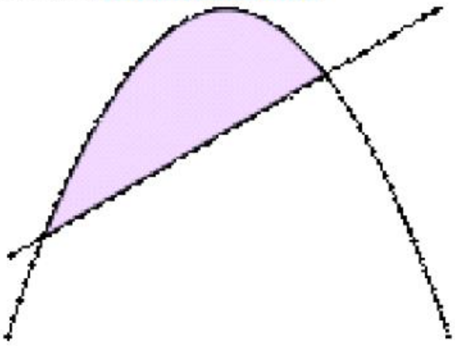
History: Ancient precursors

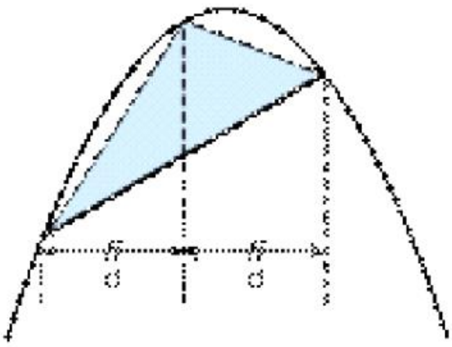
Egypt

Calculations of volume and area, one goal of integral calculus, can be found in the Egyptian Moscow papyrus (c.1820 BC), but the formulae are simple instructions, with no indication as to how they were obtained.

Greece

See also: Greek mathematics





Archimedes used the method of exhaustion to calculate the area under a parabola in his work *Quadrature of the Parabola*.

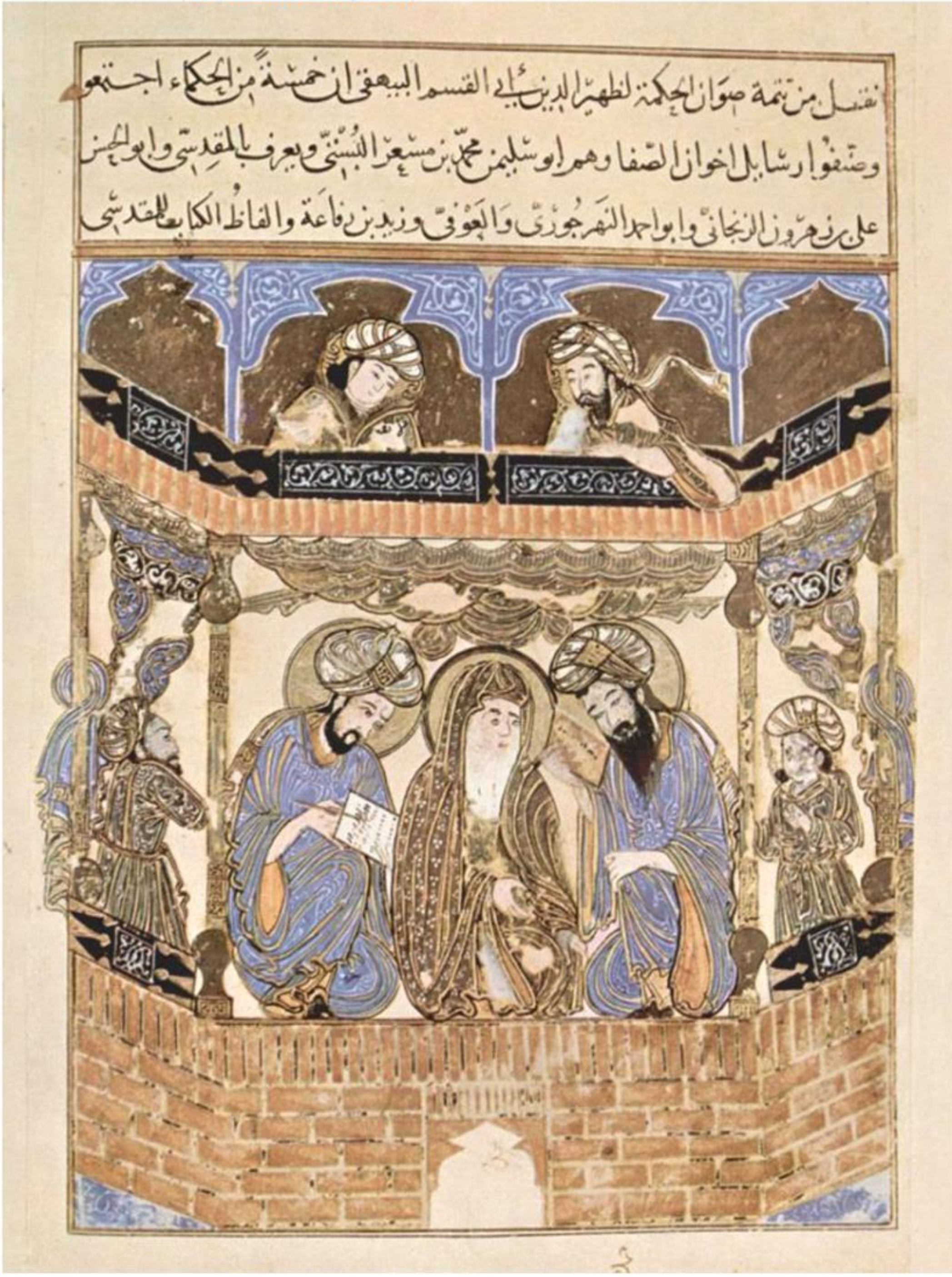
Laying the foundations for integral calculus and foreshadowing the concept of the limit, ancient Greek mathematician Eudoxus of Cnidus (c.390 – 337 BC) developed the method of exhaustion to prove the formulas for cone and pyramid volumes.

China

The **method of exhaustion** was later discovered independently in China by Liu Hui in the 3rd century AD in order to find the area of a circle.[10][11] In the 5th century AD, Zu Gengzhi, son of Zu Chongzhi, established a method[12][13] that would later be called Cavalieri's principle to find the volume of a sphere.

Middle East

Ibn al-Haytham, 11th-century Arab mathematician and physicist
In the Middle East, Hasan Ibn al-Haytham,



India

In the 14th century, Indian mathematicians gave a non-rigorous method, resembling differentiation, applicable to some trigonometric functions. Madhava of

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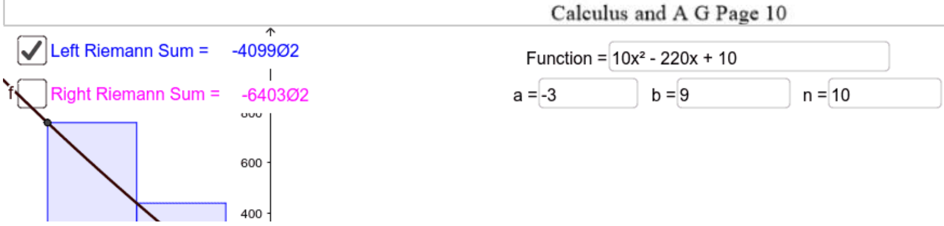
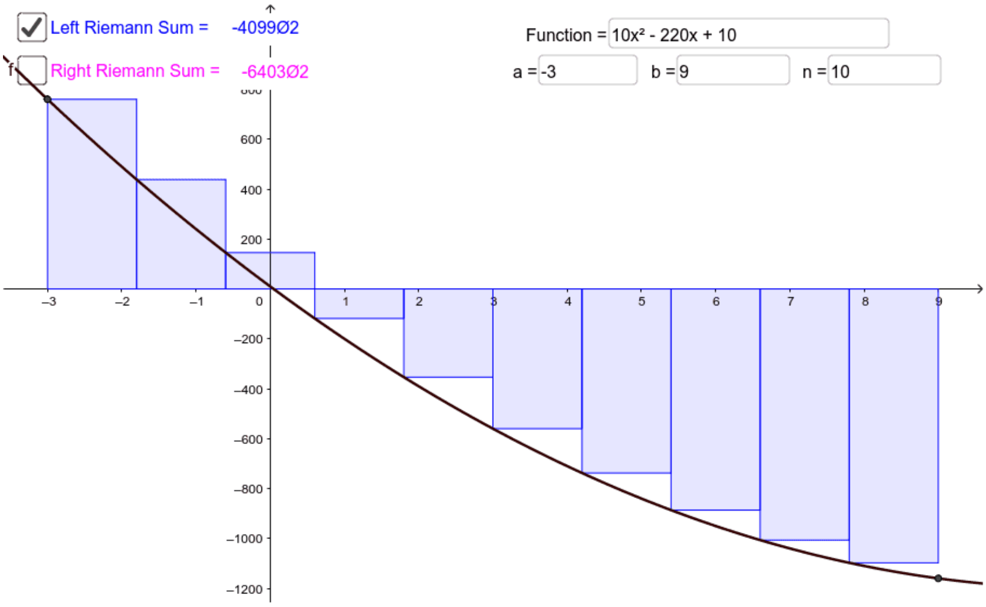
Modern Science





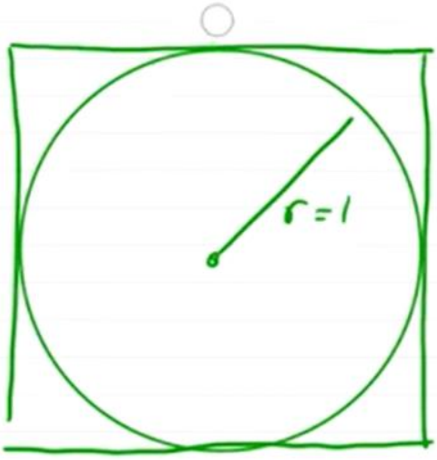
Archimedes (350 BC) Approximate π.

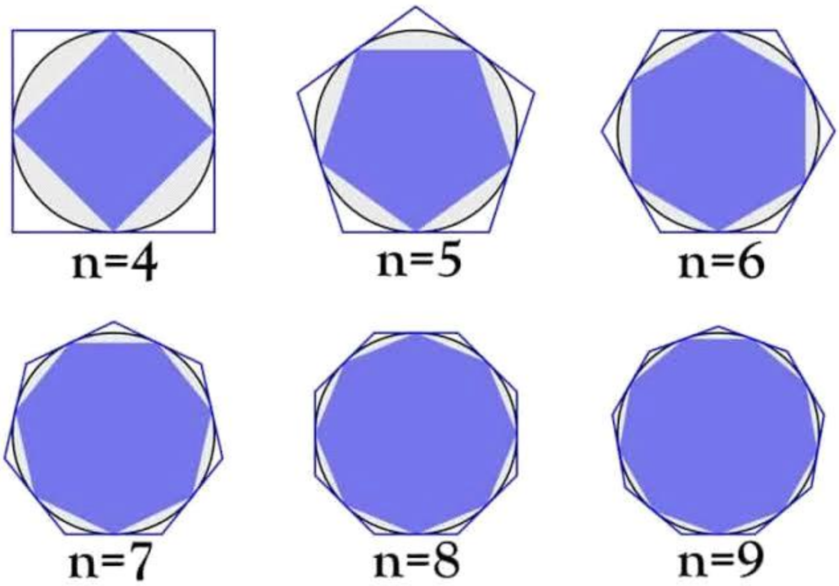
Area: $\pi r^2 = \pi$

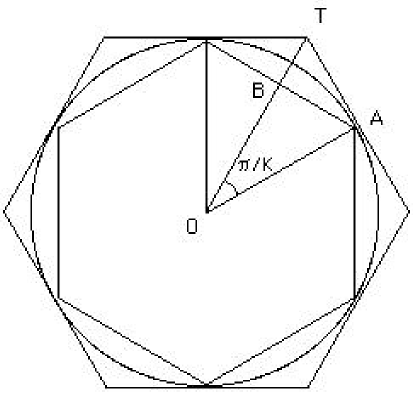
Upper bound on π is 4.

Lower bound on π is 2.

$2 < \pi < 4 \qquad \pi \approx 3.$







OA = 1
AB = sin(π/K)
AT = tan(π/K)
where K = 3 x 2^{n-1}

Heron's Method for Square Roots (1st century BC Egypt)

Goal is to find $\sqrt{5}$.

Start w/ guess $x > \sqrt{5}$. $\quad \frac{5}{x} < \sqrt{5}$

Average: $\boxed{x_{n+1} = \frac{1}{2}\left(\frac{5}{x_n} + x_n\right)}$

Logarithm Tables (Briggs Method) Henry Briggs (1561–1630)

Find log(2).

$2^{10} = 1024 \approx 10^3$

$\log(2^{10}) \approx \log(10^3) = 3$

$\log(2^{10}) = 10\log(2) \approx 3$

$\rightarrow \log(2) \approx 0.$

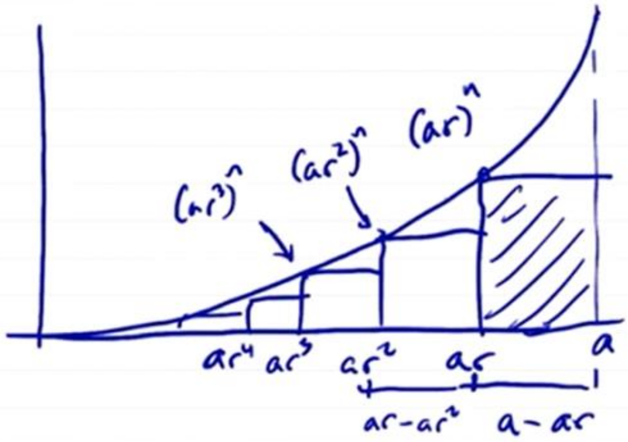
Fermat's Quadrature

Fermat (1607–1665) AP

Compute $\int_0^a x^n dx = \frac{a^{n+1}}{n+1}$

Set $0<r<1$

Geo. Series.



$$A_r = \sum_{m=1}^{\infty} (ar^m)^n \cdot (ar^{m-1} - ar^m)$$

Key concepts from Calculus

- Limits
- Continuity
- Convergence
- Differentiability
- Rolle's Theorem
- Mean Value Theorem
- Extreme Value Theorem
- Intermediate Value Theorem
- **Taylor's Theorem**

Limit / Continuity

Definition (Limit)

A function f defined on a set X of real numbers $X \subset \mathbb{R}$ has the limit L at x_0, written

$$\lim_{x \to x_0} f(x) = L$$

if given any real number $\epsilon > 0$ ($\forall \epsilon > 0$), there exists a real number $\delta > 0$ ($\exists \delta > 0$) such that $|f(x) - L| < \epsilon$, whenever $x \in X$ and $0 < |x - x_0| < \delta$.

Definition (Continuity (at a point))

Let f be a function defined on a set X of real numbers, and $x_0 \in X$. Then f is continuous at x_0 if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

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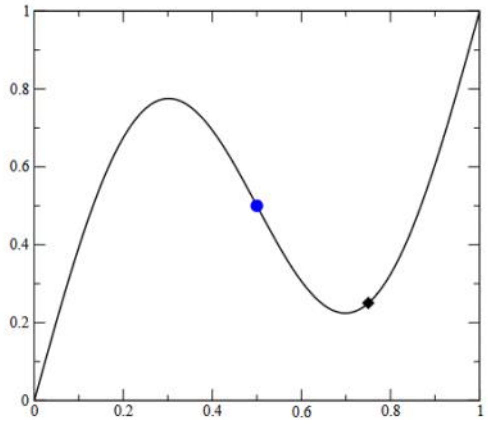
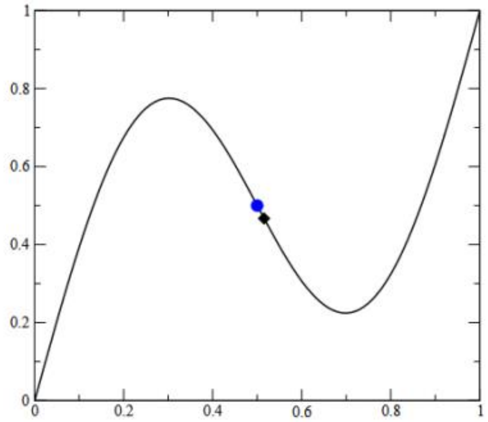
Limit Definition

Example: Continuity at x_0

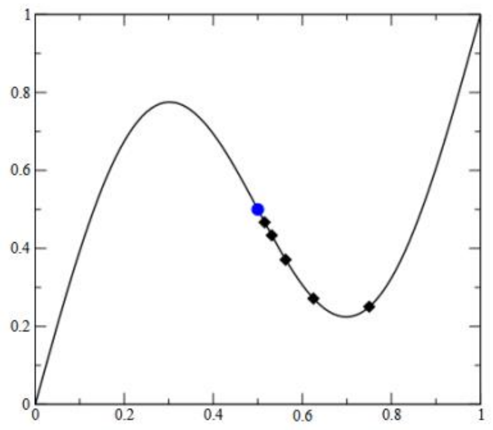


Here we see how the limit $x \to x_0$ (where $x_0 = 0.5$) exists for the function $f(x) = x + \frac{1}{2}\sin(2\pi x)$.

Example: Continuity at x_0



Example: Continuity at x_0



Concept of Continuity

1. Definition of a Limit

We say $\lim_{x \to c} f(x) = L$ if:

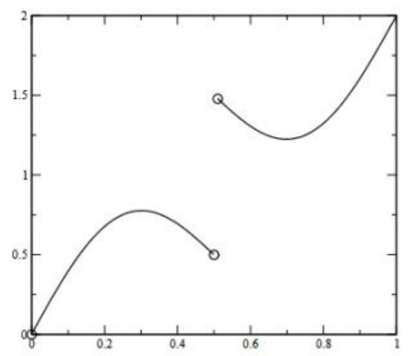
$$\forall \varepsilon > 0, \quad \exists \delta > 0 \quad \text{such that} \quad 0 < |x - c| < \delta \Longrightarrow |f(x) - L| < \varepsilon$$

2. Definition of Continuity

A function f is continuous at a point c in its domain if:

$$\forall \varepsilon > 0, \quad \exists \delta > 0 \quad \text{such that} \quad |x - c| < \delta \Longrightarrow |f(x) - f(c)| < \varepsilon$$

Examples: Jump Discontinuity

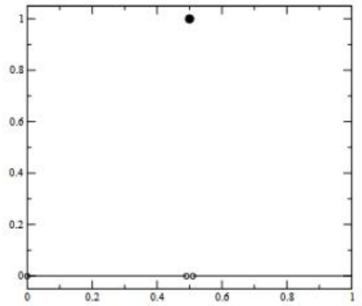


The function

$$f(x) = \begin{cases} x + \frac{1}{2}\sin(2\pi x) & x < 0.5 \\ x + \frac{1}{2}\sin(2\pi x) + 1 & x > 0.5 \end{cases}$$

has a jump discontinuity at $x_0 = 0.5$.

Examples: "Spike" Discontinuity



The function

$$f(x) = \begin{cases} 1 & x = 0.5 \\ 0 & x \neq 0.5 \end{cases}$$

has a discontinuity at $x_0 = 0.5$.

The **limit exists**, but

$$\lim_{x \to 0.5} f(x) = 0 \neq 1$$

Definition (Continuity (in an interval))

The function f is continuous on the set X ($f \in C(X)$) if it is continuous at each point x in X.

Definition (Convergence of a sequence)

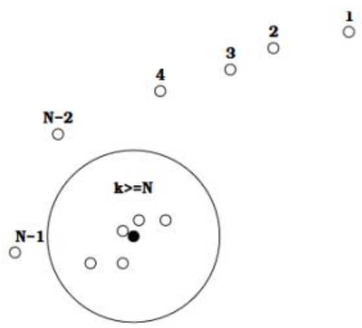
Let $\underline{\mathbf{x}} = \{x_n\}_{n=1}^{\infty}$ be an infinite sequence of real (or complex numbers). The sequence $\underline{\mathbf{x}}$ converges to x (has the limit x) if $\forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{Z}^+ \colon |x_n - x| < \epsilon \ \forall n > N(\epsilon)$. The notation

$$\lim_{n \to \infty} x_n = x$$

means that the sequence $\{x_n\}_{n=1}^{\infty}$ converges to x.

[Convergence of a sequence]

Illustration: Convergence of a Complex Sequence



A sequence in $\underline{\mathbf{z}} = \{z_k\}_{k=1}^{\infty}$ converges to $z_0 \in \mathbb{C}$ (the black dot) if for any ϵ (the radius of the circle), there is a value N (which depends on ϵ) so that the "tail" of the sequence $\underline{\mathbf{z}}_t = \{z_k\}_{k=N}^{\infty}$ is inside the circle.

Differentiability

Theorem

If f is a function defined on a set X of real numbers and $x_0 \in X$, the the following statements are **equivalent**:

(a) *f is continuous at x_0*

(b) *If $\{x_n\}_{n=1}^{\infty}$ is any sequence in X converging to x_0, then $\lim_{n\to\infty} f(x_n) = f(x_0)$.*

Definition (Differentiability (at a point))

Let f be a function defined on an open interval containing x_0 ($a < x_0 < b$). f is differentiable at x_0 if

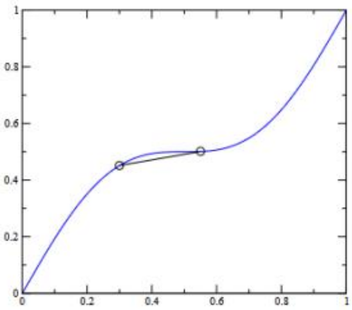
$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{exists.}$$

If the limit exists, $f'(x_0)$ is the derivative at x_0.

Definition (Differentiability (in an interval))

If $f'(x_0)$ exists $\forall x_0 \in X$, then f is differentiable on X.

Illustration: Differentiability



Here we see that the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists — and approaches the slope / derivative at x_0, $f'(x_0)$.

Illustration: Differentiability

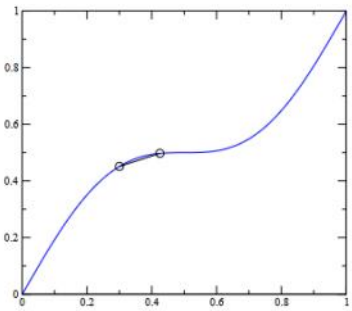


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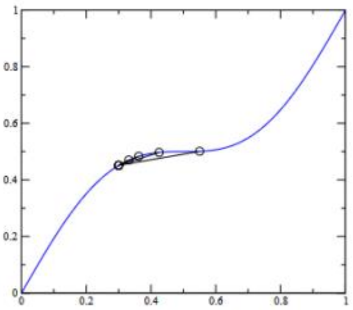
Illustration: Differentiability



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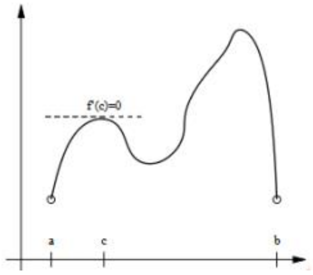
[The Definition of the Derivative]

Theorem (Differentiability ⇒ Continuity)

If f is differentiable at x_0, then f is continuous at x_0.

Theorem (Rolle's Theorem Wiki-Link)

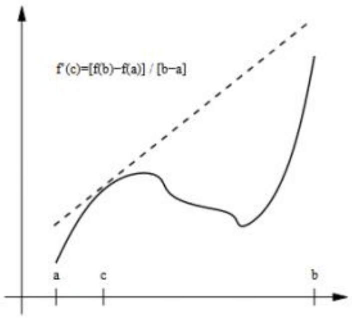
Suppose $f \in C[a, b]$ and that f is differentiable on (a, b). If $f(a) = f(b)$, then $\exists c \in (a, b)$: $f'(c) = 0$.



Mean Value Theorem

Theorem (Mean Value Theorem Wiki-Link)

If $f \in C[a,b]$ and f is differentiable on (a,b), then $\exists c \in (a,b)$: $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.

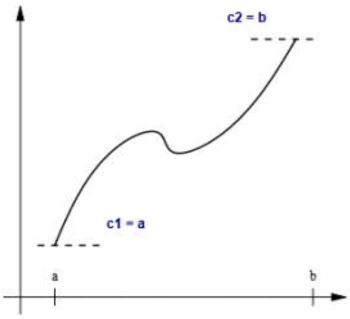


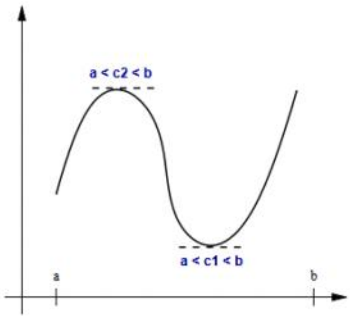
[Mean Value Theorem](#)

Extreme Value Theorem

Theorem (Extreme Value Theorem Wiki-Link)

If $f \in C[a,b]$ then $\exists c_1, c_2 \in [a,b]$: $f(c_1) \le f(x) \le f(c_2)$ $\forall x \in [a,b]$. If f is differentiable on (a,b) then the numbers c_1, c_2 occur either at the endpoints of $[a,b]$ or where $f'(x) = 0$.



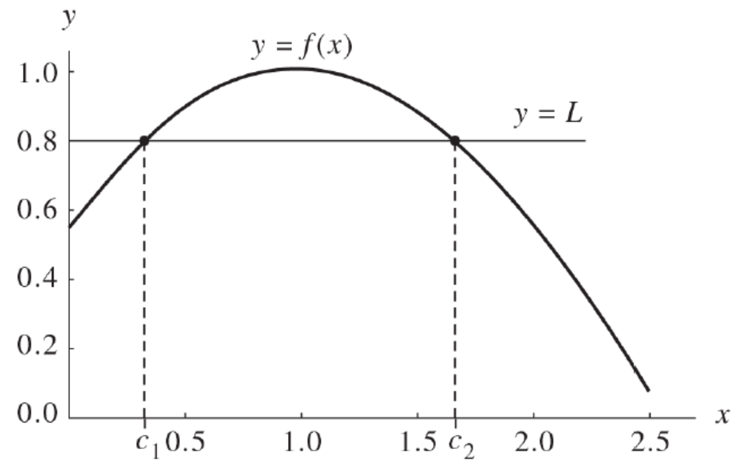


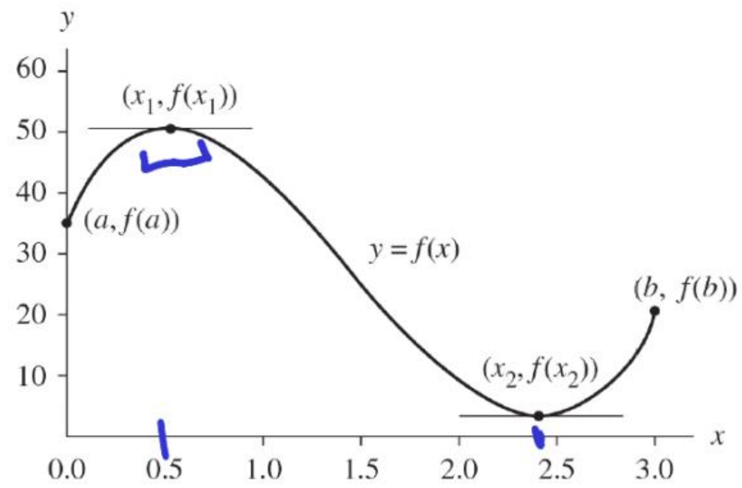
Intermediate Value Theorem

Theorem (Intermediate Value Theorem Wiki-Link)

if $f \in C[a,b]$ and K is any number between $f(a)$ and $f(b)$, then there exists a number c in (a,b) for which $f(c) = K$.

Example 1.1. The function $f(x) = \cos(x-1)$ is continuous over $[0,1]$, and the constant $L = 0.8 \in (\cos(0), \cos(1))$. The solution to $f(x) = 0.8$ over $[0,1]$ is $c_1 = 0.356499$. Similarly, $f(x)$ is continuous over $[1, 2.5]$, and $L = 0.8 \in (\cos(2.5), \cos(1))$. The solution to $f(x) = 0.8$ over $[1, 2.5]$ is $c_2 = 1.643502$. These two cases are shown in Figure 1.2. ■





Theorem 1.3 (Extreme Value Theorem for a Continuous Function). Assume that $f \in C[a, b]$. Then there exists a lower bound M_1, an upper bound M_2, and two numbers $x_1, x_2 \in [a, b]$ such that

(7) $$M_1 = f(x_1) \le f(x) \le f(x_2) = M_2 \qquad \text{whenever } x \in [a, b].$$

We sometimes express this by writing

(8) $$M_1 = f(x_1) = \min_{a \le x \le b} \{f(x)\} \quad \text{and} \quad M_2 = f(x_2) = \max_{a \le x \le b} \{f(x)\}.$$

Example 1.2. The function $f(x) = 15x^3 - 66.5x^2 + 59.5x + 35$ is differentiable on [0, 3]. The solutions to $f'(x) = 45x^2 - 123x + 59.5 = 0$ are $x_1 = 0.54955$ and $x_2 = 2.40601$. The maximum and minimum values of f on [0, 3] are:

$$\min\{f(0), f(3), f(x_1), f(x_2)\} = \min\{35, 20, 50.10438, 2.11850\} = 2.11850$$

and

$$\max\{f(0), f(3), f(x_1), f(x_2)\} = \max\{35, 20, 50.10438, 2.11850\} = 50.10438$$

(see Figure 1.3). ■

Example 1.3. The function $f(x) = \sin(x)$ is continuous on the closed interval $[0.1, 2.1]$ and differentiable on the open interval $(0.1, 2.1)$. Thus, by the mean value theorem, there is a number c such that

$$f'(c) = \frac{f(2.1) - f(0.1)}{2.1 - 0.1} = \frac{0.863209 - 0.099833}{2.1 - 0.1} = 0.381688.$$

The solution to $f'(c) = \cos(c) = 0.381688$ in the interval $(0.1, 2.1)$ is $c = 1.179174$. The graphs of $f(x)$, the secant line $y = 0.381688x + 0.099833$, and the tangent line $y = 0.381688x + 0.474215$ are shown in Figure 1.4. ■

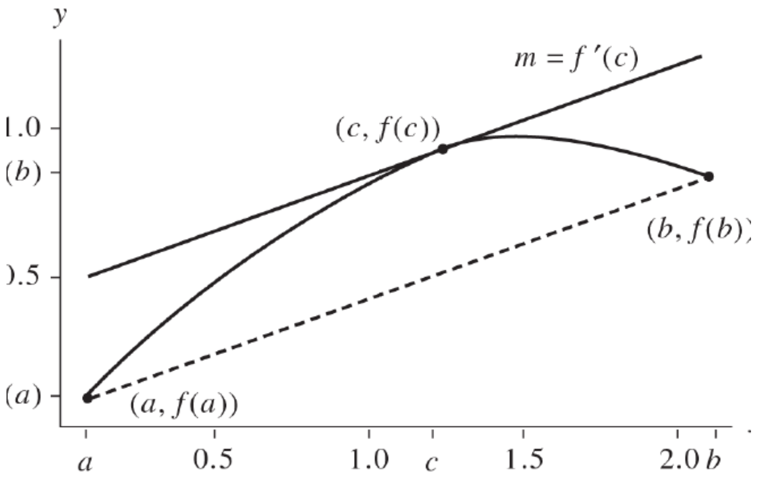


Figure 1.4 The mean value theorem applied to $f(x) = \sin(x)$ over the interval $[0.1, 2.1]$.

Integrals

Theorem 1.8 (First Fundamental Theorem). If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = \qquad \text{where } F'(x) = f(x). \tag{12}$$

Theorem 1.9 (Second Fundamental Theorem). If f is continuous over $[a, b]$ and $x \in (a, b)$, then

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x). \tag{13}$$

Example 1.4. The function $f(x) = \cos(x)$ satisfies the hypotheses of Theorem 1.9 over the interval $[0, \pi/2]$; thus by the chain rule

$$\frac{d}{dx}\int_0^{x^2} \cos(t)\,dt = \cos(x^2)(x^2)' = 2x\cos(x^2).$$ ■

Theorem 1.10 (Mean Value Theorem for Integrals). Assume that $f \in C[a, b]$. Then there exists a number c, with $c \in (a, b)$, such that

$$\frac{1}{b-a} \int_a^b f(x)\,dx = f(c).$$

The value $f(c)$ is the average value of f over the interval $[a, b]$.

Example 1.5. The function $f(x) = \sin(x) + \frac{1}{3}\sin(3x)$ satisfies the hypotheses of Theorem 1.10 over the interval $[0, 2.5]$. An antiderivative of $f(x)$ is $F(x) = -\cos(x) - \frac{1}{9}\cos(3x)$. The average value of the function $f(x)$ over the interval $[0, 2.5]$ is

$$\begin{aligned}\frac{1}{2.5-0} \int_0^{2.5} f(x)\,dx &= \frac{F(2.5) - F(0)}{2.5} = \frac{0.762629 - (-1.111111)}{2.5} \\ &= \frac{1.873740}{2.5} = 0.749496.\end{aligned}$$

There are three solutions to the equation $f(c) = 0.749496$ over the interval $[0, 2.5]$: $c_1 = 0.440566$, $c_2 = 1.268010$, and $c_3 = 1.873583$. The area of the rectangle with base $b - a = 2.5$ and height $f(c_j) = 0.749496$ is $f(c_j)(b - a) = 1.873740$. The area of the rectangle has the same numerical value as the integral of $f(x)$ taken over the interval $[0, 2.5]$. A comparison of the area under the curve $y = f(x)$ and that of the rectangle can be seen in Figure 1.5. ■

Theorem 1.11 (Weighted Integral Mean Value Theorem). Assume that $f, g \in C[a, b]$ and $g(x) \geq 0$ for $x \in [a, b]$. Then there exists a number c, with $c \in (a, b)$, such that

$$\int_a^b f(x)g(x)\,dx = f(c) \int_a^b g(x)\,dx. \tag{14}$$

Example 1.6. The functions $f(x) = \sin(x)$ and $g(x) = x^2$ satisfy the hypotheses of Theorem 1.11 over the interval $[0, \pi/2]$. Thus there exists a number c such that

$$\sin(c) = \frac{\int_0^{\pi/2} x^2 \sin(x)\,dx}{\int_0^{\pi/2} x^2\,dx} = \frac{1.14159}{1.29193} = 0.883631$$

or $c = \sin^{-1}(0.883631) = 1.08356$. ■

Series

Definition 1.5. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. Then $\sum_{n=1}^{\infty} a_n$ is an infinite series. The nth partial sum is $S_n = \sum_{k=1}^{n} a_k$. The infinite series ***converges*** if and only if the sequence $\{S_n\}_{n=1}^{\infty}$ converges to a limit S, that is,

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} a_k = S. \tag{15}$$

If a series does not converge, we say that it ***diverges***. ▲

Example 1.7. Consider the infinite sequence $\{a_n\}_{n=1}^{\infty} = \left\{\dfrac{1}{n(n+1)}\right\}_{n=1}^{\infty}$. Then the nth partial sum is

$$S_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1 - \frac{1}{n+1}.$$

Therefore, the ***sum*** of the infinite series is

$$S = \lim_{n\to\infty} Sn = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1. \qquad \blacksquare$$

1. (a) Find $L = \lim_{n\to\infty}(4n+1)/(2n+1)$. Then determine $\{\epsilon_n\} = \{L - x_n\}$ and find $\lim_{n\to\infty} \epsilon_n$.

(b) Find $L = \lim_{n\to\infty}(2n^2+6n-1)/(4n^2+2n+1)$. Then determine $\{\epsilon_n\} = \{L - x_n\}$ and find $\lim_{n\to\infty} \epsilon_n$.

2. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence such that $\lim_{n\to\infty} x_n = 2$.

(a) Find $\lim_{n\to\infty} \sin(x_n)$.

(b) Find $\lim_{n\to\infty} \ln(x_n^2)$.

For an example of a convergent sequence, let us examine $a_n = \left(1 + \frac{1}{n}\right)^n$, the well known sequence that converges to e, Euler's number.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n$

$e^{\lim_{n\to\infty} n \ln\left(1 - \frac{1}{n}\right)}$

$e^{\lim_{n\to\infty} \frac{\ln\left(1 + \frac{1}{n}\right)}{\frac{1}{n}}}$

$e^{(1)} = e$

Theorem 1.12 (Taylor's Theorem). Assume that $f \in C^{n+1}[a,b]$ and let $x_0 \in [a,b]$. Then, for every $x \in (a,b)$, there exists a number $c = c(x)$ (the value of c depends on the value of x) that lies between x_0 and x such that

$$f(x) = P_n(x) + R_n(x),$$

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}.$$

Theorem (Taylor's Theorem Wiki-Link)

Suppose $f \in C^n[a,b]$, $f^{(n+1)} \exists$ on $[a,b]$, and $x_0 \in [a,b]$. Then $\forall x \in (a,b)$, $\exists \xi(x) \in (x_0, x)$ with $f(x) = P_n(x) + R_n(x)$ where

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k, \quad R_n(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x-x_0)^{(n+1)}.$$

$P_n(x)$ is called the **Taylor polynomial of degree** *n, and $R_n(x)$ is the* **remainder term** *(truncation error).*

This theorem is **extremely important** for numerical analysis; Taylor expansion is a fundamental step in the derivation of many of the algorithms we see in this class (and in Math 693ab).

Taylor polynomial graphs

Example 1.8. The function $f(x) = \sin(x)$ satisfies the hypotheses of Theorem 1.12. The Taylor polynomial $P_n(x)$ of degree $n = 9$ expanded about $x_0 = 0$ is obtained by evaluating

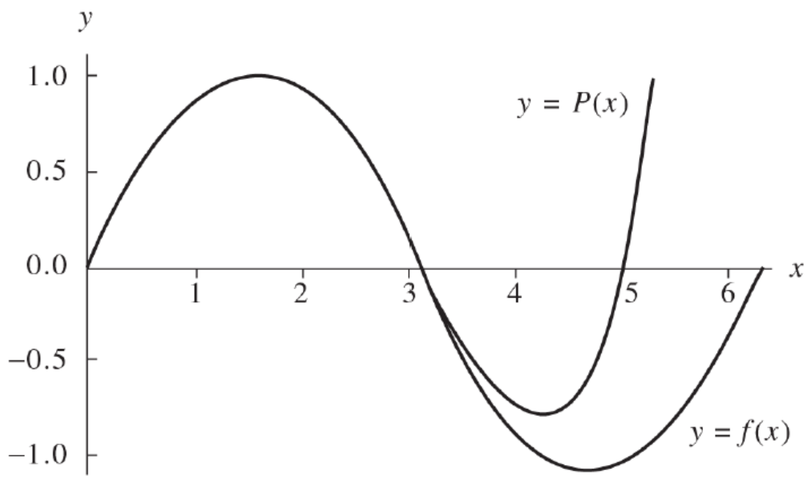


Figure 1.6 The graph of $f(x) = \sin(x)$ and the Taylor polynomial $P_9(x) = x - x^3/3! + x^5/5! - x^7/7! + x^9/9!$.

the following derivatives at $x = 0$ and substituting the numerical values into formula (17).

$$\begin{aligned}
f(x) &= \sin(x), & f(0) &= 0,\\
f'(x) &= \cos(x), & f'(0) &= 1,\\
f''(x) &= -\sin(x), & f''(0) &= 0,\\
f^{(3)}(x) &= -\cos(x), & f^{(3)}(0) &= -1,\\
&\vdots & &\vdots\\
f^{(9)}(x) &= \cos(x), & f^{(9)}(0) &= 1,
\end{aligned}$$

$$P_9(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!}.$$

A graph of both f and P_9 over the interval $[0, 2\pi]$ is shown in Figure 1.6. ■

Corollary 1.1. If $P_n(x)$ is the Taylor polynomial of degree n given in Theorem 1.12, then

(19) $$P_n^{(k)}(x_0) = f^{(k)}(x_0) \quad \text{for } k = 0, 1, \ldots, n.$$

Evaluation of a Polynomial

Let the polynomial $P(x)$ of degree n have the form

(20) $$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0.$$

Horner's method or ***synthetic division*** is a technique for evaluating polynomials. It can be thought of as nested multiplication. For example, a fifth-degree polynomial can be written in the nested multiplication form

$$P_5(x) = ((((a_5x + a_4)x + a_3)x + a_2)x + a_1)x + a_0.$$

When Horner's method is performed by hand, it is easier to write the coefficients of $P(x)$ on a line and perform the calculation $b_k = a_k + cb_{k+1}$ below a_k in a column. The format for this procedure is illustrated in Table 1.2.

Example 1.9. Use synthetic division (Horner's method) to find $P(3)$ for the polynomial

$$P(x) = x^5 - 6x^4 + 8x^3 + 8x^2 + 4x - 40.$$

	a_5	a_4	a_3	a_2	a_1	a_0
Input	1	-6	8	8	4	-40
$c = 3$		3	-9	-3	15	57
	1	-3	-1	5	19	$17 = P(3) = b_0$
	b_5	b_4	b_3	b_2	b_1	Output

Therefore, $P(3) = 17$. ■

Synthetic division is an alternative to long division. It can also be used to divide a polynomial by a possible factor, $x - k$. However, synthetic division cannot be used when the divisor is not a linear polynomial of the form $x - k$.

For example, divide $2x^4 - 5x^3 - 14x^2 + 47x - 30$ by $x - 2$.

Using synthetic division, the setup is as follows:

1. Pull down the first coefficient.

2	2	-5	-14	47	-30
	↓				
	2				

2. Multiply the coefficient by *k* and place it under the next coefficient.

2	2	-5	-14	47	-30
	↓	4			
	2				

3. *Add* the two numbers together.

2	2	-5	-14	47	-30
	↓	4			
	2	-1			

4. Repeat Steps 2 and 3 for the rest of the coefficients.

2	2	-5	-14	47	-30
	↓	4	-2	-32	30
	2	-1	-16	15	0

To "read" the answer, use the numbers as follows:

2	2	-5	-14	47	-30
	↓	4	-2	-32	30
	2	-1	-16	15	0

coefficients of factored polynomial

last number is the remainder

Therefore, 2 is a solution, because the remainder is zero. The factored polynomial is $2x^3 - x^2 - 16x + 15$.

Notice that when we synthetically divide by k, the "leftover" polynomial is one degree less than the original.

We could also write
$(x-2)(2x^3 - x^2 - 16x + 15) = 2x^4 - 5x^3 - 14x^2 + 47x - 30$.